



Norwegian University of
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Formally Verified Reduction Proof for the Free XOR Garbling Scheme

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Problem description

Garbled circuits are a fundamental cryptographic primitive used in secure two-party computation (2PC), enabling two parties to jointly compute a function without revealing their private inputs—only the output is revealed. The concept was introduced by Andrew Yao in 1986 [Yao86], where functions are expressed as Boolean circuits and each wire is labelled with encrypted wire keys. Early constructions required four ciphertexts per gate, resulting in significant communication overhead and limiting practical adoption.

Subsequent optimizations such as Point-and-Permute, Half-Gates and the Free XOR technique greatly reduced the number of ciphertexts needed per gate, as well as reducing the number of gates to be transmitted, leading to substantial performance improvements [BMR90; ZRE15; KS08]. These optimizations form the basis of many state-of-the-art secure computation frameworks.

Formal verification of cryptographic constructions is essential to ensure correctness and prevent subtle vulnerabilities arising from unchecked corner cases. Brzuska and Oechsner [BO23] developed a state-separating pen-and-paper reduction proof for Yao’s original garbling scheme. State-separating proofs model protocol components as isolated “packages” with private state and oracle interfaces, allowing for structured security reductions. Their work established security by reducing the garbling scheme to the IND-CPA security of the encryption primitive used for gate tables.

Domino is an automated verification tool specifically designed for state-separating proofs. It has been used to reason about complex real-world protocols such as TLS [Raj25] and WPA2 [BDE+22]. Domino offers a high level of automation and readability because proof scripts resemble the pseudocode found in traditional cryptographic proofs.

Furthermore, the above mentioned proof for Yao’s garbling scheme by Brzuska and Oechsner [BO23] was later formalized in Domino [BW]. It can serve as a good starting point for the thesis to observe how games may be converted from pen and paper proofs to Domino code.

The goal of this thesis is to formalize and mechanize a security reduction for the Half-Gates garbling scheme within the Domino framework. This includes:

- Developing a state-separating, pen-and-paper reduction proof for the Free XOR technique.
- Encoding this proof in Domino.

This work extends formal verification techniques to an optimized and widely deployed garbling construction and demonstrates Domino’s applicability to modern

cryptographic primitives.

Approved: 2026-03-09 – Tjerand Silde, NTNU (Main supervisor)

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of

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Preface

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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TODO: Remember to change minus to hyphen where needed.

Chapter 1

Introduction

Secure Multiparty Computation, or just Multi-Party Computation (MPC) refers to a class of problems in which two or more parties want to compute a common function using their individual inputs, but do not want to share these inputs with each other. Common use cases of MPC today include running heavy computation on cloud servers, or querying third party datasets that cannot be shared directly.

Since remote servers are often owned and operated by a separate entity, it is difficult to obtain the same security and privacy guarantees that a user's own machine can provide. Besides user requirements, cybersecurity regulations also require methods to secure the data being processed from the data processor, such as in the case of private healthcare data, or other forms of personal data [16].

Two-party computation (2PC) is a special case of MPC in which only two parties are involved in the protocol. Often attributed to Andrew Yao's seminal work in 1986 [Yao86], the first approach for 2PC was to represent the function to be computed as a boolean circuit, and to use a Garbled Circuit (GC) to compute the output of the function, while maintaining the secrecy of the inputs of both parties. For more than a decade after this work, Garbled Circuits were considered to not be practical because of the high bandwidth and computation costs of garbling, and due to the fact that garbling an arbitrary function into an optimized boolean circuit is difficult. However, there has been ongoing work in the field that has led to optimizations that allow them to be somewhat usable in generic 2PC systems like Fairplay [MNPS04], and in niche applications such as smart contracts [HYLL].

In this work, we aim to convert the proof of one such optimization (Free XOR) [KS08] to a State-separating proof (SSP) format [BDF+18], and formally verify the correctness of this converted proof. Formal verification allows us to base the correctness of a protocol or scheme on mathematical and logical constructs, rather than exhaustively testing edge cases, as is often the case in software engineering. We use Domino, a proof verification tool for SSPs, which is currently under development,

and examine the suitability of its current feature set for verifying complex proofs, while also attempting to validate the correctness of our Free XOR State-separating proof.

The initial goal of this project was to verify the Half-Gates garbling scheme [ZRE15]. However, since the Half-Gates scheme works under the Free XOR assumption, it was mandatory for us to first solve the same problem for the Free XOR scheme. Due to the limited timeframe of the project, we were unable to complete both, and leave Half-Gates as future work. We describe the details of this in further sections.

The rest of this paper is organized as follows: Section 2 contains background information on Garbling schemes, optimizations, and Domino. Section 3 details the converted pen-and-paper proof for Free XOR. Section 4 contains further details about the proof tool Domino, and an implementation of the proof, as well as comments on the tool. Section 5 contains concluding remarks and future work.

Chapter 2

Background

The following sections can be split into their own chapters (if needed).

2.1 Garbling Schemes

A Garbled Circuit can informally be defined as a two-party protocol as follows:

- steps here

How do these relate to multi party Computation

Garbling as a cryptographic goal, as opposed to a technique to achieve some specific properties.

The properties that circuit garbling provides - privacy, authenticity and obliviousness.

General definition of a garbling scheme: [BHR12]. The authors use notions of strings and functions being separate so that the garbling scheme remains generalizable and not specific to garbling boolean circuits or arithmetic circuits. Garbling scheme maps a function f to an encoding function En , a decoding function De , and an evaluation function Ev , such that when an input x is encoded by En , then the encoded value is evaluated through Ev , and the resulting output is decoded by De , the resultant value is the same as the input being evaluated by f .

Security notions definition may be included.

2.1.1 Oblivious Transfer Protocols

Why are they used in garbling schemes. This can be reduced to one line or something. Its not important to Half-Gates of Yao SSP.

2.2 Optimizations to Yao's Garbling scheme

2.2.1 Point and Permute

How many bits are supposed to be used for the wire "color"? If (x,y) are all the inputs to the circuit (if there are more than 2 inputs, x is $x_1x_2x_3...$ and similarly for

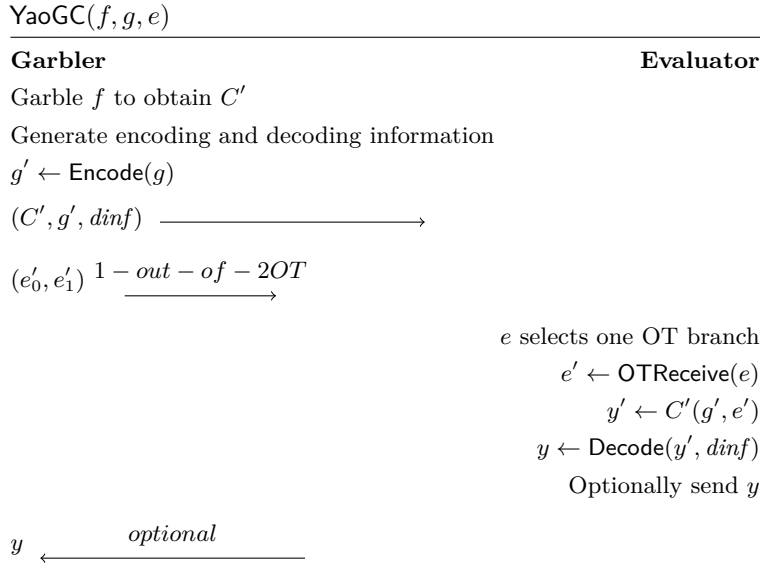


Figure 2.1: Yao's garbled-circuit protocol.

y , i.e., for each binary input label x and y , $\text{len}(x) = 2^n$ where n is the number of x inputs. similarly for y). So X and Y are the inputs domain for the two parties, and there need to be $\lceil \log(X) \rceil$ color bits after the key to preserve key length to be equal to the security parameter, while still having the ability to point to a specific X bit on the garbled truth table [EKR18].

2.2.2 Garbled Row Reduction - 3

Compatible with Free XOR

2.2.3 Free XOR

2.2.4 Garbled Row Reduction - 2

2.2.5 FleXOR

2.2.6 Half-Gates

Mention that this was the original objective of the thesis, however work with this requires freexor to be completed

2.2.7 Garbled Gadgets

2.3 Formal Verification and Tools

Start from Formal verification in general (read more too), and then go into tools used for formal verification in cryptography.

2.3.1 Easycrypt

2.3.2 Domino

How is it different from easycrypt? why did we choose this? what is a state separating proof and why is this more suited to those?

Chapter 3

Pen-and-Paper State-Separating Proof

Read half gates and chris's paper to understand what to do here even. (WIP)

3.1 State Separating Proofs

- What are games and packages
- Something about why state separating proofs exist. "composable" and "constructive" security
- State separating proofs allow us to make intermediate arguments (hybrids) in a reusable manner. It is often more intuitive to model something as its own package that has inputs and outputs than to have it be part of many different codes and realise that its the same thing.

Composability of layers is very important so we can derive circuit security from just proving layer security. The simulator for layer n must output a garbling of its output which is then supplied to a new simulator for layer $n+1$. Therefore, if a simulator can simulate a gate by getting its inputs (garbled input) and return garbled output, it can be composed with the simulator for the next gate and so on. Same idea for layers.

Actually, to add to the above, the simulator does NOT know the input. It only gets the circuit information loss of the garbled circuit $\Phi(C)$, the output of the garbled circuit Y , and the security parameter 1^λ . So it samples a random input label (in this case its the label, in the original Yao paper it was a random bitstring sampled for the wire key, but since we are working in a setting where the Free XOR optimization is assumed, we will probably end up sampling a random Δ and sharing that state across simulator invocations to generate valid labels for the wires.)

3.2 Existing Half-Gates Proof and how it is to be modelled into state separating proofs

Chapter 4

Domino

4.1 What is Domino?

What is domino and how does it work? The layers that exist, and the design in general.

Domino is a proof assistant for State-separating proofs (SSPs) that allows the user to validate a SSP using familiar syntax, and an underlying SMT solver. It uses very similar syntax and constructs as SSPs, such as packages, games and theorems. The explicit relation between these games is defined using invariant files, written in the SMT-LIB language. These games are then checked against different lemmata that are needed to fulfil the relation that the user wants to establish.

Domino is published as a rust package available on Github. The domino program/crate itself contains a compiler that has its own defined language that intentionally tries to be very similar the SSP syntax defined in [BDF+18].

4.2 Using Domino

4.2.1 Dependencies

Since Domino is published as a Rust package, it requires a Rust toolchain, which can be installed using *rustup*. Rustup manages multiple Rust toolchains, and is the most straightforward way to getting set up on Linux. Rustup is typically installed using a bash script from the official website, but it is also available on most system package managers. It can also be built from source. Installing rustup automatically installs the Rust binary and cargo (the package manager), among other components. These can also be obtained without rustup, and the installation page explains the methods in more detail [Rus].

The second prerequisite to run Domino is a functioning CVC5 installation, and for the cvc5 executable to be added to PATH, so that Domino can invoke it while

running. CVC5 can be built from source following the instructions in [CVC]. CVC5 is the SMT solver that checks whether the constraints given by our proof are satisfiable or not. This is done in practice by taking the negation of the condition we want, and checking for satisfiability of that negation. If it exists, the condition is false.

4.2.2 Project Structure

A Domino project directory requires the presence of an empty *ssp.toml* file. It allows SSP logic to be written into packages, games and theorem files. These are all placed under their own directories, i.e., folders called *packages*, *games*, and *theorem*.

Each of these folders can then contain one or more files. Domino checks packages and games for syntactic correctness even if they are not used in a theorem. This is intentional design, since when writing state separating proofs, the components (packages and games) are usually set up before the actual proof.

In older versions of Domino, theorems were called proofs, and the folder and file header for proofs were named as such.

An Example Theorem in Domino

mIND-CPA game [here](#), talk about its packages, and the two games with diagrams, and how the theorem to prove states that the two games are indistinguishable.

4.3 Coding Half-gates in Domino

4.4 Improvements

Chapter 5

Conclusion

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